SOCIAL SCIENCE
BAL VIGYAN
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Last, but not the least we would like to thank our parents who directly or indirectly helped us in completing the project.

DISSERTATION APPROVAL SHEET

This is to certify that the presented dissertation entitled "**Seeds For Change**" has been completed by Ms. Parnika Budholiya, Ma. Arghya Sharma, Ma. Arham Pahadia and Ms. Aadhya Jain of "The Shishukunj International School, Jhalaria" under the guidance of the Social Science Department.

I wish them success in their work.

Dr. Lalita Singh (Principal)

The Shishukunj International School

INTRODUCTION

“Agriculture is the most healthful, most useful, most noble employment of a man”-George Washington

Agriculture is the backbone of our civilization and our economy. Almost 2/3rd of India’s population is involved in some form of agricultural activity. However, have we ever thought about the condition of people who work day and night to produce grains for the country?

Poverty in the Agricultural sector remains a big issue in our nation. Almost 60% of the total population of our nation is employed in the agricultural sector but still the contribution of this sector to our economy is only 16%. By the implementation of new technological innovations the agricultural sector can increase drastically contributing massively to our GDP.

The Indian government provides many subsidies and benefits to Indian Farmers. However, a survey held by our students concluded that the majority of the small/marginalized farmers are not at all aware about these schemes made by the government. For an example, almost 51% of the total farmers in India are still not aware about the Pradhan Mantri Fasal Bima Yojana which was introduced by our government so as to provide financial support to farmers suffering crop loss/damage arising out of unforeseen events and for stabilizing the income of farmers to ensure their continuance in farming.

Madhya Pradesh is a state where almost 70% of the population is either directly or indirectly dependent on agriculture to sustain their livelihood. In a state such as Madhya Pradesh it becomes very important to come up with methods which help the farmers to increase their incomes.

Education is a very important aspect of the same. Farmers of Madhya Pradesh are comparatively less educated than the farmers of states such as Kerala and this often leads to exploitation of farmers by private industries.

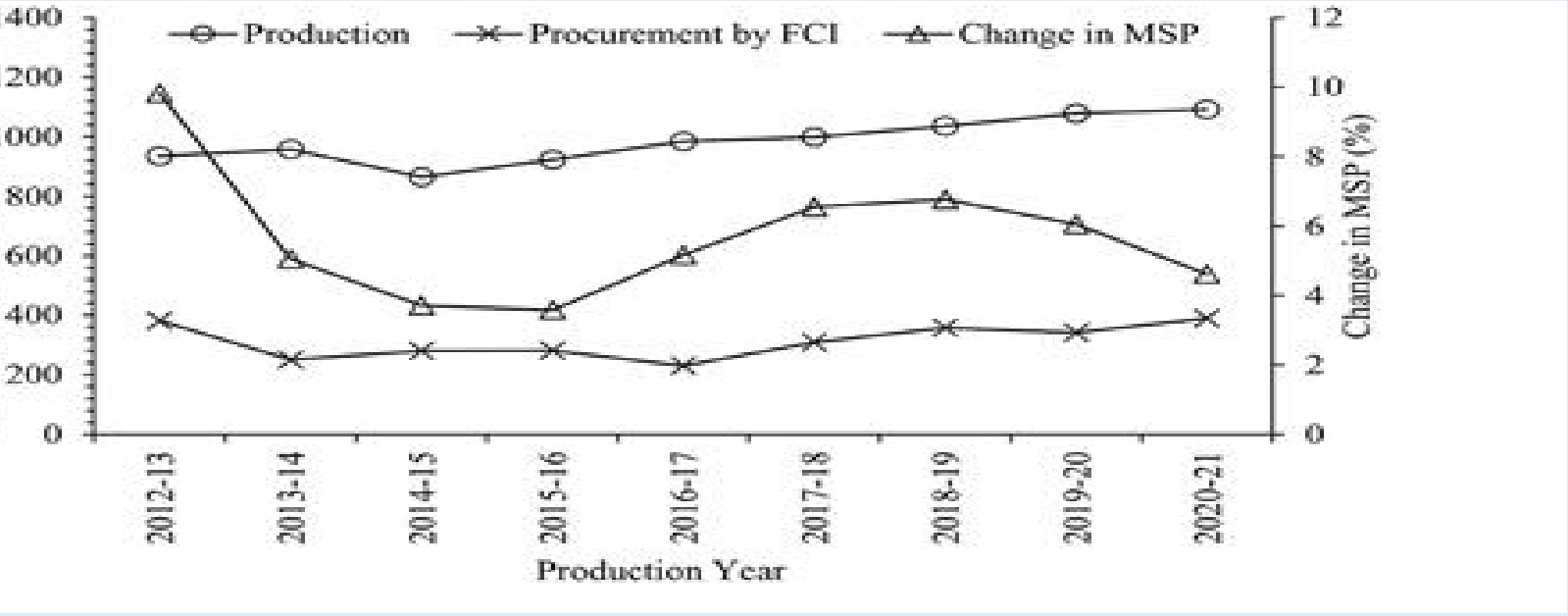
One might believe that farmers of our country are economically stable and well to do because of the large number of consumers present in our country. However, a report by the NSSO (National Sample Survey Organization) shows that an average farmer earns only around 74,000 rupees per year. That is the case for an average farmer, the condition of poor farmers is way worse than what can be imagined.

Small farmers in India, Madhya Pradesh to be more precise face a variety of problems ranging from economic instability and lack of resources to natural factors such as inconsistent rainfall.

IDENTIFICATION OF THE PROBLEM

The major problems faced by our farmers are: inadequate irrigation facilities, no proper electricity source, high prices for agricultural inputs (raw materials), and a lack of proper maintenance and storage of crops.

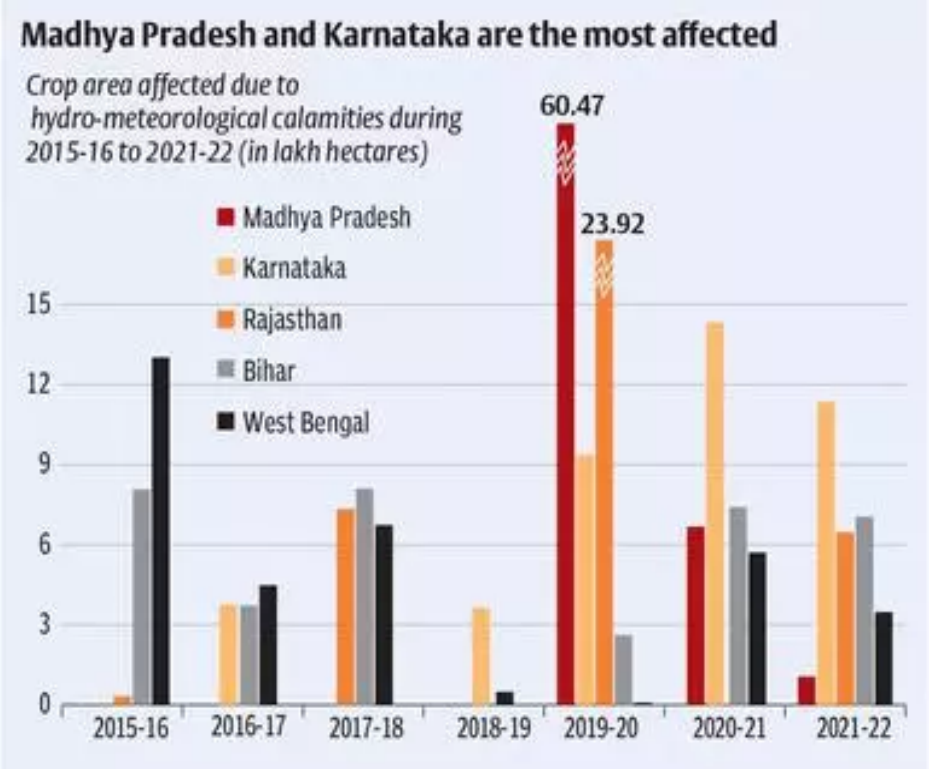
The inconsistent Minimum Support Price offered by the government isn't helping farmers procure proper raw materials and machinery, which leads to low productivity.



Furthermore, we live in a society where political interests and corruption are at times kept higher than civil interests, here is a case study that proves the inaccurate implementation of the government schemes.

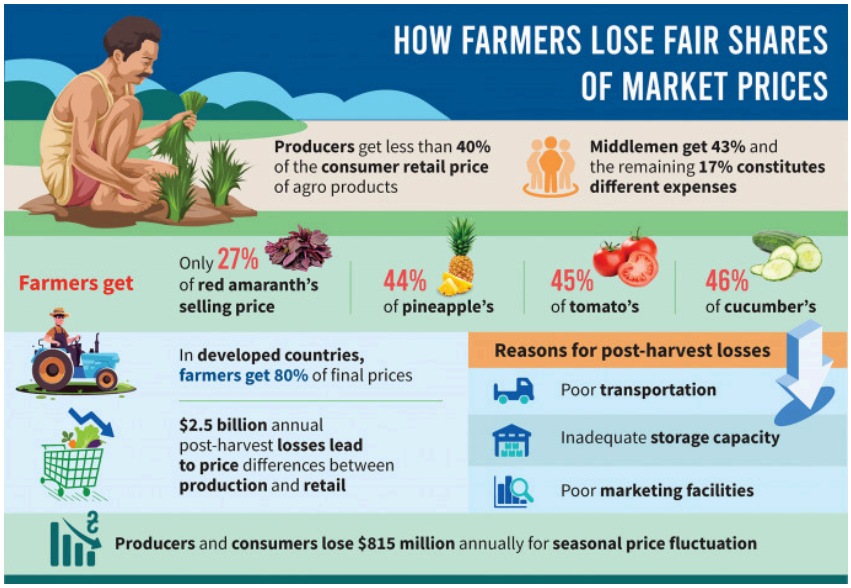
Damaged soybean crops and meagre compensation for farmers despite the much-hyped Prime Minister's crop insurance scheme figure prominently in the complaints of farmers at Chhapri, a picturesque village near Budhni, about 70 km south of Bhopal, which is abuzz with political activity.

Patiram Bhalla, a peasant, said that the government's minimum support prices are an eyewash. "In 2020-21, jowar [sorghum] crops were weighed at the government mandis for procurement, but 2,500 farmers in Budhni are yet to be paid." Mukesh Kangal complained that procurement by the government is often delayed, forcing most farmers to sell their produce at throwaway rates to private bidders.



An example of the same is of 2019 when The state was expecting a bumper harvest with estimates of wheat production at upwards of 20 million tonnes. Unseasonal rains in mid to late March have adversely impacted the crop estimates but they have also delayed the harvesting. About 80 percent of the wheat crop in MP is harvested using combine harvesters. There are a total of 4,500 harvesters locally registered in MP but they are not enough to harvest the entire wheat crop. The support comes from the combine harvesters that come from Punjab every year. The peak harvesting dates in MP precedes peak harvest season in Punjab and other north Indian states in most years. But the delays with the lockdown mean MP has to manage on its own this year. This is just an example of how climate change affects various fields of agriculture such as harvesting, sowing, growing and e.t.c. This makes it very clear that farmers of Madhya Pradesh suffer through a shortage of very basic needs, adding to which Madhya Pradesh's growing unstable climate hampers their growth. Due to all of these factors Madhya Pradesh is now one of the most inconsistent states in terms of production of grains. Which means that farmers of Madhya Pradesh have no security and surety about their future incomes.

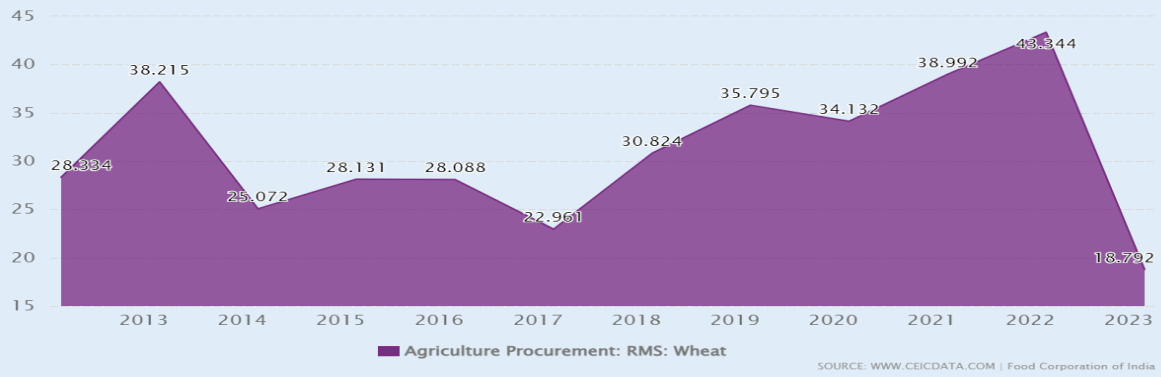
Cold storage facilities are one such key agriculture infrastructure aspect. Madhya Pradesh has 250 cold storages, out of which only five cold storages, barely two percent of the overall cold storages in the state, are govt-run. Out of these five government-run cold storages also, three are dedicated to dairy products, one is for potatoes and only one is for the remaining crops.



This lack of education is often used as a source of benefit by the private sector companies who are involved in the field of agriculture. A case study by our students highlights this issue-

The cotton farmers of Madhya Pradesh's Nimar region are known to be a prosperous lot. This year, however, they are feeling betrayed by the seed companies that sold them Bt cotton. Nimar, which includes the districts of Khandwa, Burhanpur, Khargone, Badwani and Dhar, succumbed to the promise of `Bt', which was promoted aggressively as a product that would lead to high yields with lower investments, and was suitable for all types of soil and water conditions. At least three lakh seed packets were sold in the region, covering at least 22 per cent of its 6.34 lakh hectares of cotton-sown area. Unfortunately, the farmers were not warned about the fine print at the bottom edge of the packets. While the posters and pamphlets loudly promised gains in both irrigated and non-irrigated land. This led to a year of great loss for the farmers.

The struggle is not over yet, our farmers work day and night in order to produce grains however many times they do not get appropriate prices for the crops. This is because of the high rates of money which is taken by the middle man. A report shows that producers get only less than 40% of the total worth of crops which they produce. Majority of which is consumed by the middle man This is a serious issue in our country because of which farmers remain poor.



The lack of workers at procurement centers and the scarcity of HDPE bags needed for the procedure would slow down MP's purchase process in the interim. This year, the state government is employing more than just jute bags; it is combining jute and HDPE bags. The state government has bags that can hold around 6.4 million tonnes of wheat, and underground storage silos have the potential to hold an additional 900,000 tonnes of wheat. To handle around 4 million tons, the state administration has made orders to purchase additional HDPE bags. However, it is a tall order to expect the bags to reach in time, given that factories are closed as a result of the lockdown and transportation is severely impacted.

Another issue which is a major threat to farmer's welfare is the social divisions created amongst the farmers themselves. Social divisions have made differences in the farmer community which restricts the flow of knowledge and awareness among the farmers. The lack of farmer unification also limits farmers' movements to protest against government policy.

Farmers also face the issue of the lack of knowledge regarding new agricultural practices and techniques as they often stick to the traditional farming methods and practices. They are reluctant to change their production processes and are in a vicious cycle of traditional occupational stigma. An example of this is the cultivation of mushrooms. Generational farmers are not very open to the idea of growing and selling mushrooms, because they have not been made aware of the ease of its production and the economic benefits that it reaps. They are also not aware of the mandi system for the selling of mushrooms.
This lack of knowledge also threatens the the enterprising of the famer's production. Without knowledge of the newer techniques and practices, farmers are unable to access machineries that connect the rural and urban sectors for production. This connectivity produces more information for the farmer for new innovative practices to reap economic benefits. The lack of this connectivity results in the restriction of their entrepreneurship.

AIMS

The main objectives of this project are -

1. **<u>Improving the quality of life of farmers by introducing them to crops they can grow alongside their main crops.</u> This will help them with:**

a. Getting more profits

i. This allows them to live a better life and get more material. A major problem farmers face is the lack of equipment which is due to the low revenue of farmers. After they adopt our plans, they can get a considerable increase of revenues which shall allow them to get more tractors, better seeds, better fertilisers and more equipment.

b. Procuring capital

i. As the income of a farmer increases, they can put more money as capital. This again helps them with the problem of lack of equipment.

c. **Motivating more people to pick up farming as a profession which will lead to India being self sufficient**

i. As people see that farmers are gaining more and more profits, they will be attracted to the industry. This will help our country as a whole in becoming a major exporter of agricultural and other related products. This will also attract foreign industries to our country as they will get cheap raw material.

2. **<u>Educating farmers about various new schemes and technologies</u> which leads to :**

a. **Awareness about government schemes and new schemes**

i. The government uses around 1,20,500 crore Rupees on government subsidies yearly but these are not used by the farmers as they do not know or do not understand it. By the virtue of this fact we feel the need for awareness.

b. **Efficient farming thus maximizing profits**

i. As the farmers get to know about subsidies and new technology the farmers will be able to maximize their profits.

c. Modernization of Indian Farming Industry

i. Indian agriculture is still very primitive when compared to western agriculture. We, through our efforts, want farmers to adopt newer technology.

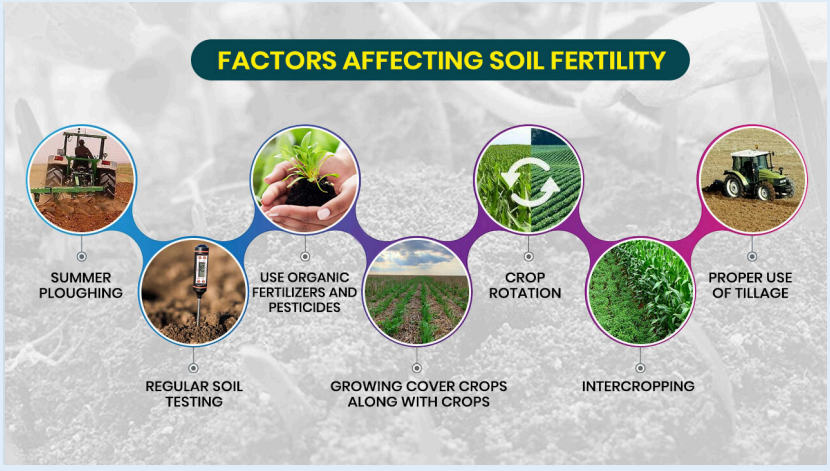
d. Awareness about sustainable-farming

i. We want farmers to be aware about the problems that intensive usage of fertilizers can cause and also adopt sustainable farming which will help replenish the soil quality.

3. <u>Promoting sustainable-farming</u> which :

a. Improves overall land quality each year, thus improving yield

i. Due to intensive use of fertilizers, pesticides and overwatering the overall soil quality of agricultural land has reduced. This reduces the yield given by land. Sustainable farming will help them maximum yield from any given land.



b. Maintains the ecological balance

i. While certain organism's harm plants, some of them are important for the growth of plants. Overuse of pesticides, insecticides and other poisons disturbs the ecological balance of farms. This again reduces the yield of farms. Hence telling them about ecological balance of a farm land is one of our goals.

c. Reduces the need for chemical fertilizers every year

i. The main problem with chemical fertilizers is that they rapidly degrade the soil. Thus making them aware about the problems of these will reduce their usage thus helping the soil recover.

4. <u>Promoting the social connections between small farmers</u>. This will help them with

a. Getting the required equipment

i. Promoting the formation of SHGs which will allow pooling of equipment and thus reducing the overall cost of practice

b. Being united this forming a lucrative vote bank

i. Uniting the farmers into SHGs will allow them to act as one united group which will allow them to sway the government.

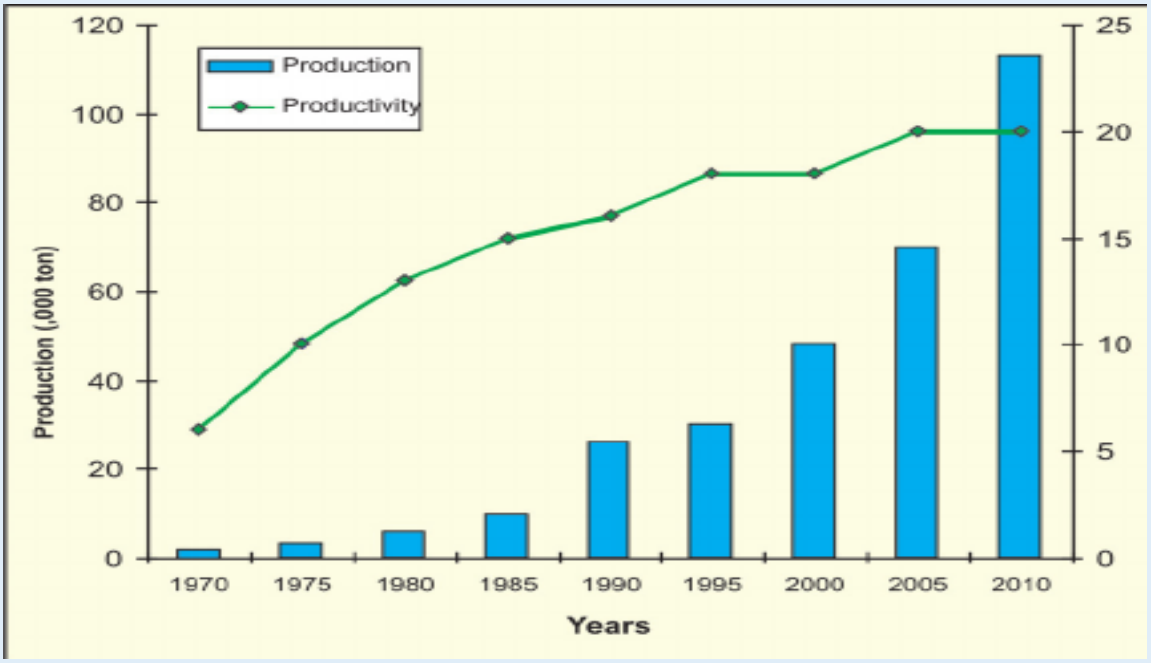
c. Connecting to markets

i. A major problem faced by farmers is that they grow produce but are not connected to the market. This also demotivates them from exploring newer, more profitable crops. Thus we want to connect them to the markets where they could sell their produce.

d. Exchange of Information amongst farmers

i. By forming Kaissan Jagrukta Sabha's (KJS)we will start the exchange of information amongst farmers. This will also allow flow of information and experiences between them. This will prevent farmers from making wrong decision

RESEARCH AND DATA COLLECTION



PRODUCTIVITY OF MUSHROOMS OVER THE YEARS

The fresh mushroom segment is likely to continue dominating the global mushroom market. The segment accounted for over 89.5% of total mushroom market revenue in 2021. Fresh mushrooms have seen significant growth in demand in recent years as a result of an increased number of consumers focused on organic and unprocessed food consumption to reap the most health advantages from the product. Due to their short shelf life, fresh form distribution is difficult for producers and distributors. However, advances in technology, such as the use of modified environment packaging, have reduced the obstacles associated with mushrooms' short shelf life, supporting segment expansion.

HERBAL MEDICINE MARKET GROWTH IN INDIA

India's significant position in the global herbal medicines market, standing at a robust 42%, underscores its pivotal role in shaping the industry's landscape. The anticipated growth trajectory, with an estimated market size of $2.6 billion by 2027 and a compound annual growth rate (CAGR) of 11.7% during 2022-2027, reaffirms India's expanding foothold in the herbal extracts sector.

This growth isn't just about numbers; it mirrors a growing global appreciation for traditional herbal remedies and India's prowess in harnessing its rich botanical heritage. As the demand for natural, plant-based solutions continues to surge worldwide, India stands as a major contributor to meeting this demand, offering diverse herbal extracts renowned for their medicinal and therapeutic properties.

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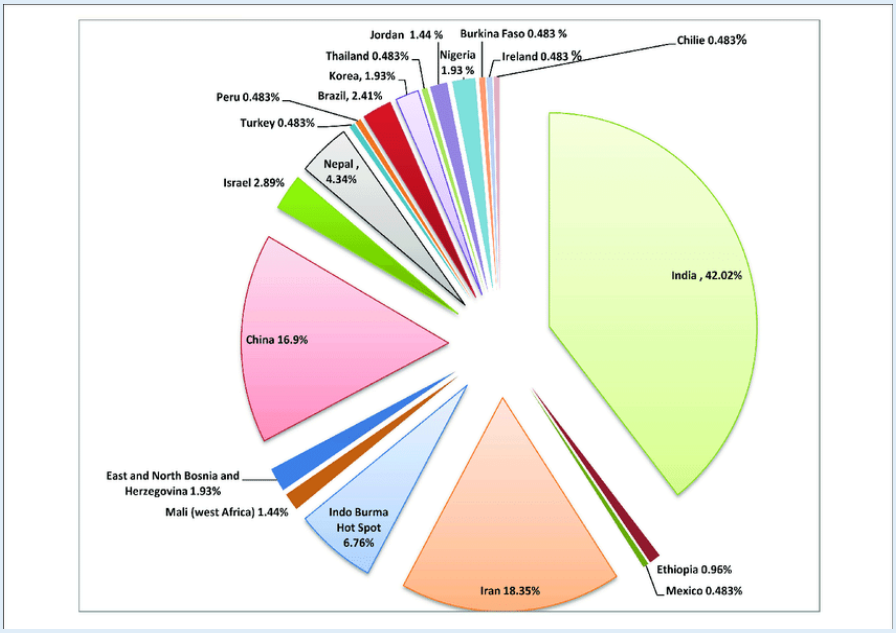
The expanding market presents opportunities for various stakeholders, from traditional medicine practitioners to modern pharmaceutical companies, to explore and harness India's wealth of herbal resources. This growth trajectory can potentially foster advancements in research, production, and global distribution networks, further solidifying India's position as a leader in the herbal extracts market.
India's pivotal role isn't just confined to market figures; it symbolizes a heritage deeply rooted in holistic healing traditions that are increasingly gaining recognition and acceptance on a global scale.



SHARE OF INDIA IN GLOBAL MUSHROOM SALE

India's mushroom sector has indeed experienced robust growth over the analyzed 25-year period (1997-2021), leveraging conducive factors such as abundant resources and a burgeoning domestic demand. The utilization of Compounded Annual Growth Rate (CAGR) to scrutinize production, import, and export patterns sheds light on the sector's evolution.

The culmination of this trend in 2021, marked by a significant production output of 243 metric tons, highlights the sector's resilience and upward trajectory. This production milestone signifies a substantial increase compared to earlier years, underscoring the sector's ability to expand and meet the escalating demand within the country. The thriving nature of India's mushroom industry can be attributed to various factors, including technological advancements, increased awareness about the nutritional benefits of mushrooms, favorable climatic conditions

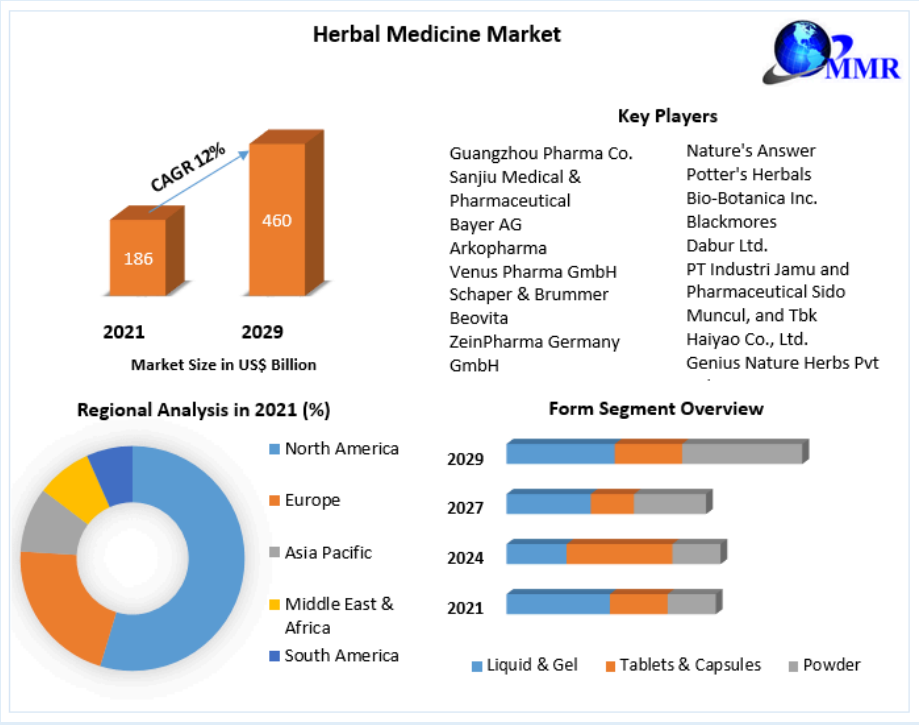
in certain regions, and the proactive involvement of both government and private entities in promoting mushroom cultivation.

Moreover, while the domestic demand acts as a strong driving force for the sector's growth, it's also worth noting that the export potential of Indian mushrooms has garnered attention in international markets. The quality, variety, and volume of mushroom production in India have made it a prospective player in the global market, presenting opportunities for export expansion.

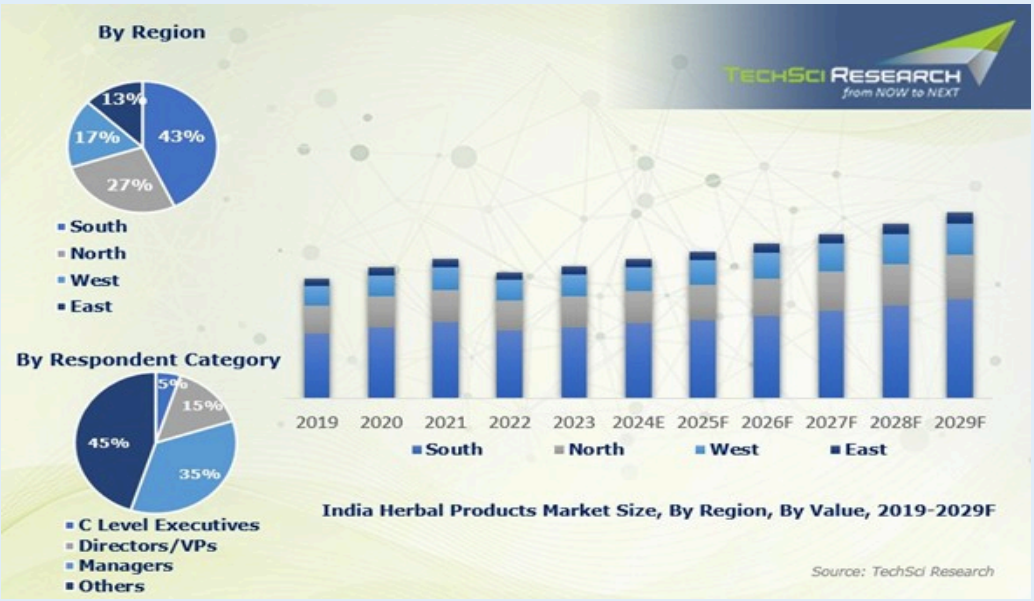
The utilization of CAGR as an analytical tool unveils not just the growth patterns but also underscores the sector's potential for sustained development in the coming years. As the mushroom sector continues to capitalize on these conducive factors and embraces advancements in cultivation techniques and marketing strategies, it's poised for further growth, both domestically and on the global stage.

EXPANSION OF HERBAL MARKET:

Due to the global resurgence in traditional and alternative healthcare systems, the market for herbal drugs has grown at an impressive rate. The global market for herbal drugs is lucrative and therefore the world herbal trade is expected to reach USD 7 trillion by 2050. The Indian share of this world trade at present is quite low although the advancements within the country are booming. Total global herbal market is of size 62.0 billion dollars, in this India's contribution is only one billion dollars.



The market for herbal medicine in India has been steadily expanding due to increasing awareness of natural remedies and a growing preference for holistic healthcare. Factors like cultural acceptance, traditional knowledge, and government support for Ayurveda contribute to this growth.



India is currently one of the top exporters of alternative medicines worldwide. India is positioned to be a worldwide leader and is expected to experience an outstanding 50% increase over the course of the next five years thanks to a strong foundation in ancient Ayurveda and the large use of ayurvedic medications.

Indian herbal market is registering extremely significant growth and is likely to reach Rs.14,500 crore (Rs 145,000 million) by 2012 and exports to Rs.9,000 crore (Rs 90,000 million) with a CAGR of 20% and 25% respectively, according to findings of the Associated Chambers of Commerce and Industry of India (Assocham).India estimated about 860 billion worth export of herbal raw material and formulated drugs to various developed countries. The cultivation of herbal medicine is one of the most profitable agribusiness for any farmer in India and requires moderate investment. For example, a farmer cultivating atis, an herb largely utilized in Ayurvedic medication, can easily earn ₹2.5-3 lakh per acre of land. Madhya Pradesh also ranks first as far as production of herbal medicines is concerned.

Along with this, the National Medicinal Plants Board (NMPB) offers up to 75% subsidy to farmers across the country who are getting this subsidy from the AYUSH ministry to empower the cultivation of medicinal plants as part of a centrally initiated government scheme.

SURVEYS:

Bijalpur:

Surveyee: Jitu Jirati (Ex MLA)
Omprakash Arya (Zila Panchayat Board and Farmer)

1. What are some of the agricultural problems you face in your locality?
आपके इलाक़े में कुछ कृषि संबंधित समस्याए क्या है?

Ans: The major problems faced by our farmers are: irrigational facilities , no proper electricity source , high price for agricultural input (raw materials) and lack of proper maintenance and storage of crops.

2. During the Zila Panchayat Meet , what challenges and problems are faced by the farmers all over the sector?
ज़िला पंचायत की बैठक में आप कौन कौनसी चुनौतियाँ और समस्याओं से अपने किसानों को जूझते हुए देखते है?

Ans: The Minimum Support Price offered by the government isn't enough to procure proper raw materials and machinery , leading to low productivity. Also , crops are sometimes thrown away due to bad crop turnout.

3. How do you ensure that all farmers in your locality stay updated on the latest schemes and ideas?
आप कैसे सुनिश्चित करते है कि आपके इलाक़े के सारे किसान नवीनतम योजनाओं और विचारो से जागरूक रहे?

Ans: Firstly , the companies themselves come for the product endorsement once in a while and secondly , the local bank managers come to the village and promote awareness in regard with governmental schemes and subsidies; however , they agree there is no proper system of raising awareness for all.

4. Are programmes such as Krishi Darshan made available and are regularly updated for your farmers?
क्या कृषि दर्शन जैसे टेलीविज़न कार्यक्रम किसानों के लिए उपलब्ध करवाये जाते है?

Ans: Yes these programmes are available however they aren’t suitable and localised to provide for the needs of the farmers and so , they aren't the most reliable source for our farmers.

5. How do you inform the farmers of the weather conditions and rainfall for the upcoming season?
आप अपने किसानों को मौसम की स्थिति और वर्षा की स्तिथियों के परिवर्तन से जागरूक कैसे करवाते है?

Ans: Since our farmers have been practising agriculture for years , they are accustomed to the weather conditions and change their practices according to their needs.

6. Do you have any newspapers or pamphlets to raise awareness about the new technology and resources?
क्या आप किसी भी प्रकार के समाचार पत्रों या पुस्तिकाओं का वितरण करते हैं?

Ans: No , however we would appreciate weekly reports for education and bettering the situation of our farmers.

7. What do you feel about the non conventional methods of farming? e.g Hydroponics which is soil less farming etc.
पारंपरिक तकनीकोंसे थोड़ी हटकर जो कृषि अभ्यास तकनीकिया होती है उनके बारे में आपके क्या विचार है?

Ans: Although we appreciate the idea and realize that it will increase our profit but at the same time , looking at the present situations , we don't think it is viable for our farmers considering the cost , time and effort required to achieve this fully.

8. Do you think villages will be willing to switch to these methods?
क्या आपको लगता है कि सारे गाँव इन तरीको पर बदलने के लिए इच्छुक है?

Ans: Yes , if proper funding and training is provided to our farmers on a regular basis.

9. Will your farmers be willing to come together regularly to discuss new methods and innovations regarding farming?
क्या आपके किसान इन नहीं योजनाओं और तकनीको के बारे में सोच-विचार करने के लिए नियमित रूप से सम्मिलित होना चाहेंगे?

Ans: We have a Krishi Samiti at present however it isn't regular , so we would like the setup of a committee for regular meetings , to be headed by a government official to ensure efficiency.

10. What services are provided to ensure financial aid to your farmers so that they do not fall into debt traps and have sufficient money for procuring resources?
आपके किसानों को एसी कौनसी वित्तीय सहायता से संबंधित सेवाए प्रदान की जाती है जिससे कि वह कृष्ण जाल में ना फसे?

Ans: We do have the help of Seva Sahakari Sanstha and some nationalised banks however it isn't the most easy and viable option for our farmers so we would like the setup of some more local banks or systems to help educate and help our farmers when in need.

11. What all changes and techniques would you like to implement to better the conditions of your farmers and enhance the crop yield?
आप कौन कौनसे बदलाव और तकनीको का इस्तेमाल देखना चाहेंगे की किसानों की फसल में सुधार दिखे?

Ans: Firstly,we would like some schemes to be organised by the government specifically for the small farmers with the lands ranging from 2-3 hectares. Since we have continuous problems of lack of proper irrigation and electricity , we would like the government to provide sprinkler subsidies free of cost as well as fixed electricity units which would be looked after by the government .Lastly , we would also like the set up of a 'farmers committee' which would hold regular meetings , headed by a government official to better tackle and provide for the needs of our farmers.

Kanadia:

1. For how many years have you been engaged in farming?

आप कितने वर्षों से खेती कर रहे हैं?

Ans: Most of us have engaged in farming since we were young.

2. Do you at times feel like there are insufficient resources for practicing agriculture? And if so ,then what all problems do you face ?

क्या आपको कभी-कभी ऐसा लगता है कि कृषि कार्य करने के लिए संसाधन अपर्याप्त हैं? और यदि हां, तो आपको किन-किन समस्याओं का सामना करना पड़ता है?

Ans: Yes, a lot of times our crops do not come out as expected. We believe it is because we are not able to get high end fertilizers and pesticides to increase crop yield and improve quality.

3. Do you know about some new technological advancements , for example solar irrigation,automation systems ?

क्या आप कुछ नई तकनीकी प्रगति के बारे में जानते हैं?

Ans: No, no one tells us about new technology.

4. How often does the Gram Panchayat of your locality update you on the new best techniques for farming?

आपके इलाके की ग्राम पंचायत आपको खेती की नई सर्वोत्तम तकनीकों के बारे में जानकारी दे
है?

Ans: The Panchayat sends some people to tell us about medicines for the crops but these visits aren't frequent and the medicines are also not cheap.

5. Are you aware of any govt schemes made specially for farmers? And if yes, what schemes and policies do you personally benefit from? (tell about KCC, beej yojana, subsidies on hydroponics)

क्या आप किसानों के लिए विशेष रूप से बनाई गई किसी सरकारी योजना के बारे में जानते हैं?

Ans: Some of them, but we are not able to avail them because we are not told how to and where to reach out for getting the benefits.

6. Which all major crops do you sow?

आप कौन सी प्रमुख फसलें बोते हैं?

Ans: We grow Wheat, Soya, Chickpeas and Corn.

7. Do you know that there are more profitable crops that you could grow to increase your income, for example mushrooms , lettuce etc.

क्या आप जानते हैं कि ऐसी अधिक लाभदायक फसलें हैं जिन्हें आप उगाकर अपनी आय बढ़ा सकते हैं, उदाहरण के लिए मशरूम, सलाद आदि?

Ans: We do not know much about mushroom's(answer of majority of the farmers) but one of them had formerly practised mushroom farming (Mr. Radhakrisna Mandloi) and gave a positive response.

8. Horticulture is in trend right now and its production rate is very high . Would you consider switching ?

बागवानी इस समय प्रचलन में है और इसकी उत्पादन दर बहुत अधिक है। क्या आप स्विच करने पर विचार करेंगे?

Ans: We would welcome any practice that would help us in getting more profit but you will have to prove it to us first.

9. Once you harvest, where do you go to sell your crop?

एक बार जब आप फसल काट लेते हैं, तो आप अपनी फसल बेचने कहां जाते हैं?

Ans: We sell it in the sabzi mandi in local Indore.

10. What is the process of you selling the crop? Who are the middlemen involved and are they of any help?

आपकी फसल बेचने की प्रक्रिया क्या है? इसमें कौन बिचौलिए शामिल हैं और क्या वे कोई मदद करते हैं?

Ans: Since we directly sell our harvest in the Mandi there are not many middlemen involved in our sale.

11. At what rates do you get to sell your crop at? And do you think you would get better rates without the interference of the middlemen?

आपको अपनी फसल किस रेट पर बेचने को मिलती है? और क्या आपको लगता है कि बिचौलियों के हस्तक्षेप के बिना आपको बेहतर दरें मिलेंगी?

Ans: We get around ₹2500 per quintal for wheat and around ₹50 per kg for Soya.

12. What information do you have on the current mandi and msp system? Do you think it is beneficial? If not, what would you want changed?

वर्तमान मंडी और एमएसपी प्रणाली पर आपके पास क्या जानकारी है? क्या आपको लगता है कि यह फायदेमंद है? यदि नहीं, तो आप क्या परिवर्तन चाहेंगे?

Ans: We get information about the mandi and MSP however we are not frequently made aware about it.

Anu Tech mushrooms-

1)Has any work done for farmers welfare in the form of mushrooms up till now?

Ans: We avail the small farmers of sources for growing mushrooms in the lowest start capital and the highest income on call also. Sources such as mushroom seeds and spawn at low cost, chemicals that give quick results.

2)Have you indulged in any mushroom growing kit programmes?

Ans: Yes, mushroom growing kits for 100 bags are available at low cost in ₹1,500 to ₹2,000. Along with this mushrooms grow best in the winter season.

3)Would you like to indulge in mushroom cultivation and technology specifically in Kanadia and other agricultural areas near Indore?

Ans: Yes, of course, we can host a 2 days camp with all the sources and facilities, we prefer growing the oyster variety of the mushroom. The first product was bought by us.

4)Do you connect to any NGO for supplying mushroom growing kits and training to the small/marginalized farmers in a moderate way?

Ans: No, we provide these mushroom growing kits directly to these farmers.

5)Would you like to subsidize the rate of training to small/marginalized farmers?

Ans: Absolutely! The cost reduces and the maximum with the resources is 3,500 whereas the cost changes from the no. of farmers.

6)What has been the response of the small-scale farmers on mushroom training?

Ans: 6 out of 10 small-scale farmers are willing to undergo the mushroom training provided by us.

7)What technology has been taught to the farmers and how many farmers had joined the last session?

Ans: Chemical based technology for oyster mushrooms is used as they are easy to grow and the price for production doesn't increase. The mushrooms require hot water to sterilise for 1 hour.

8)What all drawbacks do you face when you educate the farmers of your programmes and schemes?

Ans: The farmers see more work and less income for e.g. for 100 bags the average income after 1 month is ₹10,000 which demotivates the farmers but they have to realise that everything starts small and as they learn they are able to generate more income with the less investment start-up work and less resources thus saving money.

9)In the future, what programmes and schemes would you like to develop to better the facilities for the farmers?

Ans: We are contacting NGOs in areas such as Manawar and Burhanpur, NGOs are active and they work equally for everyone and thus we have meetings with these NGOs. Tying up with the NGOs and giving them the startup capital and innovation in order to ultimately increase the mushroom rowers all over India.

OUR SOLUTIONS

Keeping in mind our utmost commitment to farmer's welfare and the gravity of the problems faced by the small and marginal farmers today, we have put in our tremendous efforts for the betterment of the farmer at the grassroot level. We have the proposition of the following solutions for the maximization of the economic benefit and upgradation of the production techniques:

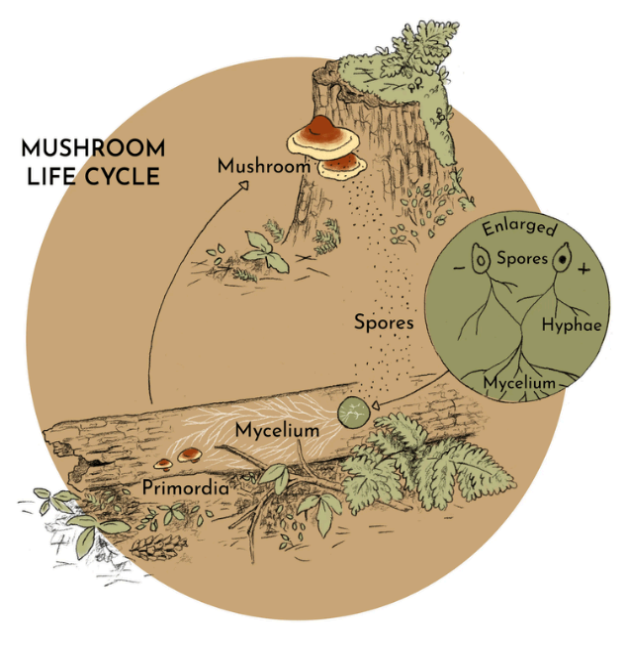
1. CULTIVATION OF MUSHROOMS

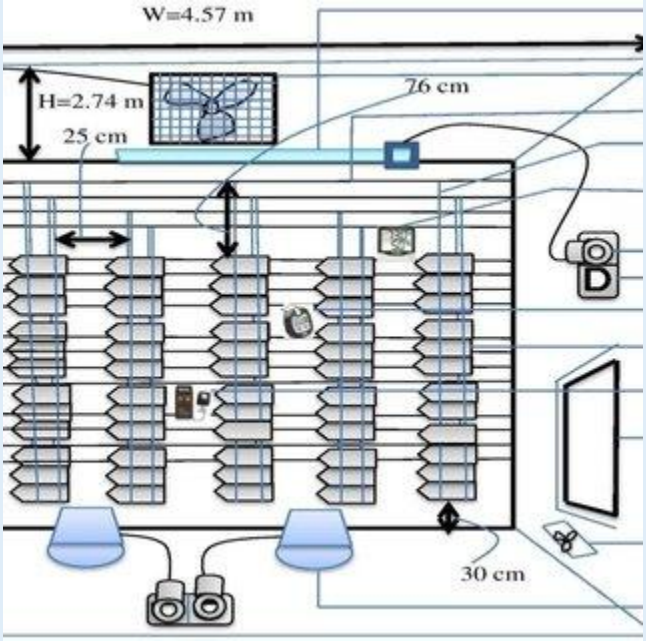
We plan on cultivating a plot of mushrooms on land contributed to us by the generous farmers of the Kanadia village. This will be done under the guidance of Mr. Mandloi, an ex- mushroom farmer in the village of Kanadia. We also have Mr Shasheesh Shrivastava, a resource expert on mushroom cultivation from Jabalpur. To assist us on the innovations and the technical aptitude of mushroom farming.

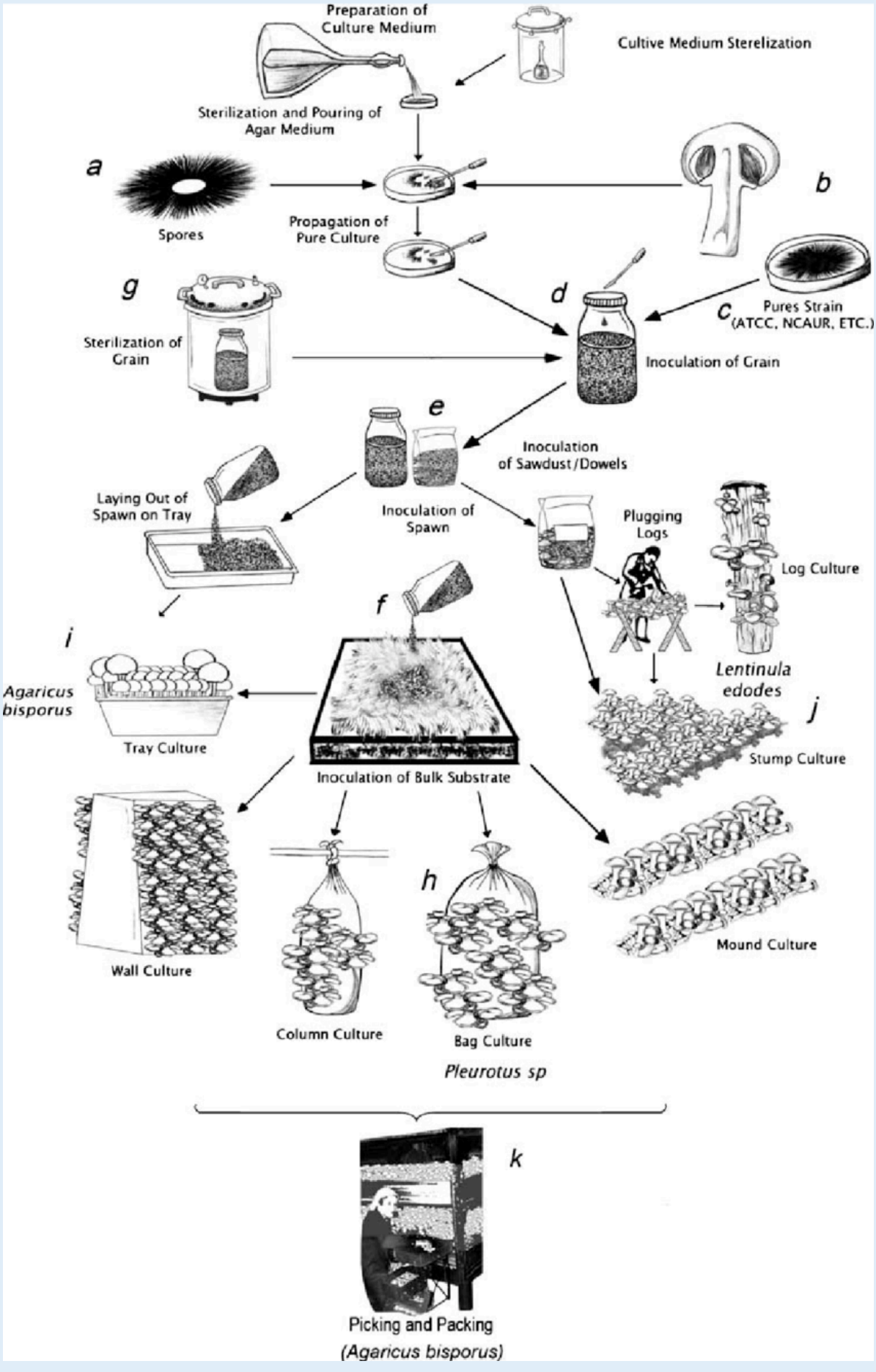
The demo plot for the mushroom cultivation will be a small closed room as mushrooms do not require sunlight to grow. It is hung in gunny bags and grows only under special conditions at 75 percent relative humidity and at a temperature less than 20 degrees. Mushrooms start growing even naturally when the humidity is five percent, so mushrooms are grown indoors.f there is little air conditioning then air cooling can be done by installing a cooler.

The production of oyster mushrooms, also known as blue lotus, will be promoted as they have proven to be the most profitable and easy to grow for small and marginal farmers. It can be grown by soaking wheat straw in formalin and Bavistin water and leaving it overnight. The wheat straw is easily available with the farmers or if bought, is purchased at Rs. 5 to 8 per kg. One kilo of straw turns into half a kilo of mushroom and one kilo of mushrooms is produced per square foot of area. A minimum of 2,500 kg and a maximum of 5000 kg of fresh mushrooms can be extracted and the production cost of mushrooms today is around ₹ 50 per kg. Whereas it sells for an average of Rs. 125 per kg. A small and marginal farmer can earn 200% profit in this, if all the conditions remain perfect.

The government has set up and is running many projects. For example, the Government of India has a big institute on mushrooms in Chamba, Solan, where you can get training on button mushrooms, oyster mushrooms and others. After training, bank loans also become easy. There is also a back end subsidy of the Horticulture Department, which sometimes ranges from 20 to 40 percent and goes upto 50 percent. This can be applied to the National Horticulture Board.






Preparation of Culture Medium
Cultive Medium Sterelization
Sterilization and Pouring of Agar Medium
a
Spores
Propagation of Pure Culture
b
g
Sterilization of Grain
d
Inoculation of Grain
c
Pures Strain (ATCC, NCAUR, ETC.)
e
Inoculation of Sawdust/Dowels
Laying Out of Spawn on Tray
Inoculation of Spawn
Plugging Logs
Log Culture
f
i
Agaricus bisporus
Tray Culture
Lentinula edodes
j
Stump Culture
Inoculation of Bulk Substrate
Wall Culture
Column Culture
h
Bag Culture
Mound Culture
Pleurotus sp
k
Picking and Packing
(Agaricus bisporus)

MUSHROOM CULTIVATION IN MINES

Mushroom cultivation within abandoned mines signifies a remarkable fusion of agriculture and unconventional spaces. The process commences with a meticulous selection of suitable mines, accounting for crucial environmental factors like temperature, humidity, and ventilation. Upon identifying optimal locations, extensive cleaning and sterilization procedures are undertaken to prepare the mine for cultivation. Mushroom spores or mycelium are then introduced into a nutrient-rich substrate, often composed of organic materials such as straw and wood chips. This mixture is methodically spread across designated cultivation beds within the mine, fostering a conducive environment for mushroom growth. The mine's sheltered and controlled conditions offer an ideal space, safeguarding the mushrooms during their developmental stages.

Moreover, this innovative agricultural practice isn't limited solely to mushrooms. Open-shaft mines, for instance, can also be repurposed for cultivating lotus flowers, presenting an additional avenue for economic gain. The flowers not only flourish in such conditions but also yield marketable roots, diversifying the revenue potential from this unconventional farming method.

By harnessing abandoned mines for agricultural purposes, this approach embodies resourcefulness and sustainability. It transforms otherwise neglected spaces into productive hubs, offering an eco-friendly alternative to traditional farming practices. This ingenious method optimizes these spaces, effectively utilizing their unique environmental characteristics to yield valuable crops and contribute to sustainable agricultural practices in an innovative manner.

2. AYURVEDIC FARMING

Ayurvedic farming, employing organic methodologies, offers a multitude of benefits, fostering environmental sustainability and human well-being. The practice encompasses organic techniques like companion planting, crop rotation, and employing plant-based solutions for natural pest control. Key to this approach is soil enrichment through composting and vermicomposting, vital for bolstering soil fertility, water retention, and fostering microbial activity.

Crucially, the reliance on organic methods obviates the necessity for expensive synthetic inputs, effectively reducing production costs. This shift empowers small-scale farmers to curtail external input expenses by crafting their own natural fertilizers and pesticides from locally available resources. Furthermore, it opens doors to a burgeoning niche market specifically for Ayurvedic plants, presenting farmers with new avenues to expand their agricultural pursuits.

By delving into Ayurvedic farming with organic techniques, farmers not only embrace eco-friendly practices but also tap into a lucrative market for Ayurvedic produce. This shift not only aligns with sustainable agricultural principles but also equips farmers with the potential to substantially augment their income, offering a pathway to greater economic stability and growth.

Organic Ayurvedic farming prioritizes biodiversity and sustainability. Farmers focus on preserving indigenous plant varieties, promoting biodiversity, and preserving traditional knowledge. This approach not only sustains ecological balance but also safeguards rare and valuable plant species essential for Ayurvedic medicine. By avoiding synthetic chemicals, farmers preserve the integrity and potency of medicinal plants, ensuring their efficacy for health and wellness purposes.

Additionally, Ayurvedic farming's emphasis on organic principles contributes to environmental conservation. By eschewing chemical fertilizers and pesticides, this method minimizes soil and water contamination, safeguarding natural resources and fostering healthier ecosystems. It promotes the cultivation of medicinal herbs in their natural habitat, enhancing their medicinal properties and contributing to the preservation of natural flora and fauna.

Moreover, the cultivation and commercialization of Ayurvedic plants through organic farming methods align with global trends favoring natural and sustainable products. This paradigm shift towards organic and environmentally friendly practices positions Ayurvedic farming as an integral part of the broader movement toward conscious consumerism and holistic wellness. It not only meets the rising demand for natural remedies but also promotes a holistic lifestyle while preserving the environment for future generations

3. INTRODUCTION TO SOILLESS FARMING

Hydroponics and aeroponics are innovative soilless farming methods. Hydroponics involves growing plants in nutrient-rich water solutions, while aeroponics suspends plants in air and delivers nutrients through mist. Both techniques optimize resource use, providing controlled environments for efficient and accelerated plant growth, revolutionizing traditional agriculture.

HYDROPONICS

Hydroponic farming represents an innovative and resource-efficient method of cultivation, where plants grow without soil, drawing essential nutrients directly from water fortified with mineral solutions. This soil-less technique involves various systems like the nutrient film technique (NFT), deep water culture (DWC), or vertical farming setups.

The method offers several advantages, including optimal resource utilization—requiring less water compared to traditional soil-based farming—and the ability to grow crops in limited spaces, making it suitable for urban environments.

Hydroponics enables precise control over nutrient levels, pH balance, and environmental factors, fostering accelerated plant growth and higher yields throughout the year regardless of external conditions.

Moreover, this technique minimizes the risk of soil-borne diseases and pests, reducing the need for pesticides and herbicides. Hydroponic systems can be tailored for specific crops, maximizing efficiency and productivity. While initial setup costs might be higher, the long-term benefits in terms of consistent yields, efficient resource utilization, and reduced environmental impact make hydroponic farming an increasingly viable and sustainable option for modern agriculture.

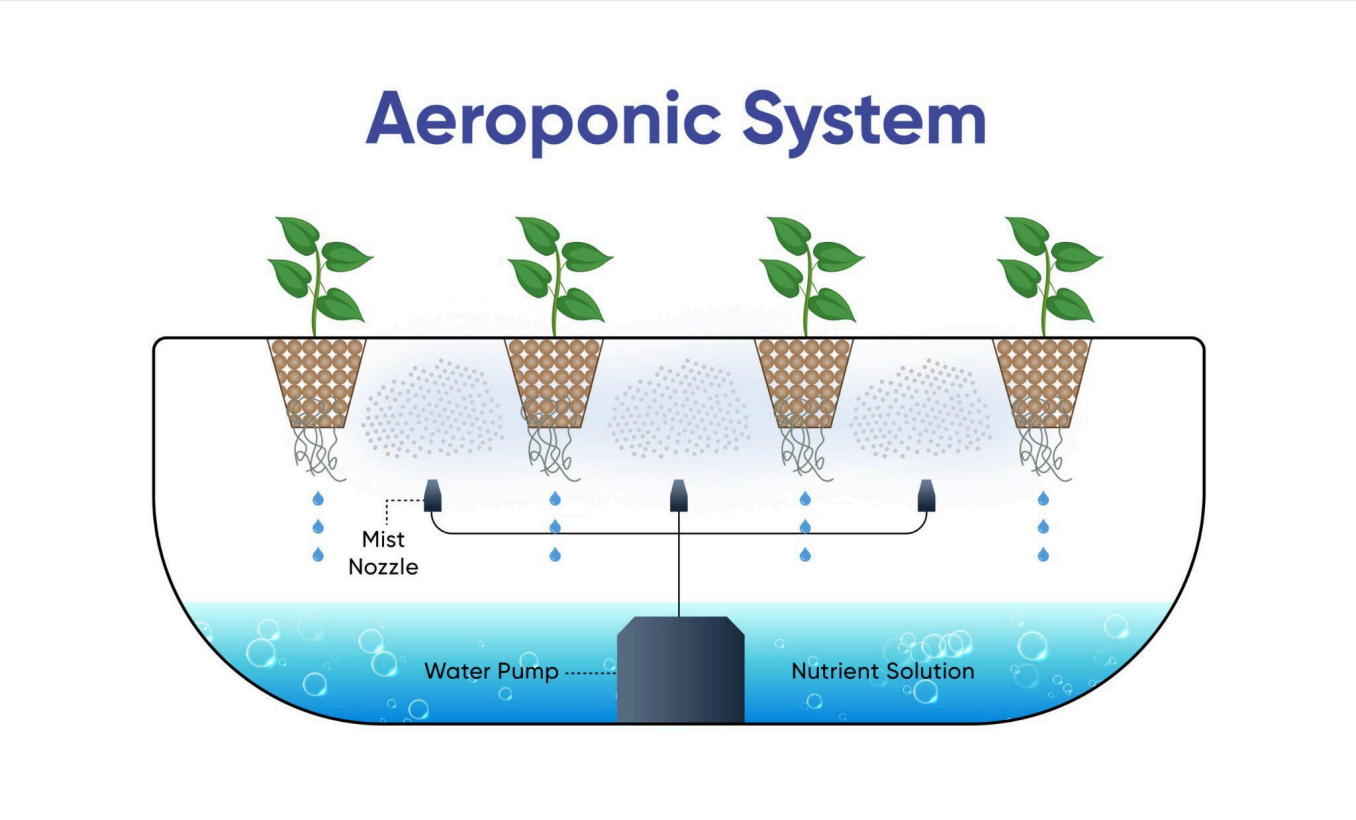
AEROPONICS

Aeroponics stands as an advanced and soil-less method of cultivating plants, where roots are suspended in the air within a controlled environment. Unlike hydroponics, which uses water as the medium for delivering nutrients, aeroponics employs a mist or air to disperse nutrients directly onto the plant roots.

This technique offers several advantages, primarily efficient nutrient absorption by the plants as the roots are exposed to a rich mist of nutrients and oxygen. With a precisely controlled environment, including nutrient concentration, pH levels, and humidity, aeroponics allows for accelerated plant growth and higher yields compared to traditional methods. Additionally, it requires significantly less water than conventional farming and even hydroponics, making it a more resource-efficient option.

Aeroponics also minimizes the risk of diseases and pests associated with soil-based cultivation, reducing the need for pesticides. The method is adaptable and suitable for various plants, including herbs, leafy greens, and certain vegetables. Despite requiring initial investment in specialized equipment and maintenance, the long-term benefits

in terms of superior plant growth, increased yields, and resource efficiency position aeroponics as a promising and sustainable farming technique for the future.



Furthermore, the precise control over the plant's environment in aeroponics facilitates the optimization of growth conditions. This means that factors such as light exposure, temperature, and CO2 levels can be finely tuned to meet the specific needs of different plant species or growth stages. By tailoring these variables, growers can achieve ideal conditions for maximizing photosynthesis and overall plant health. This level of control also enables year-round cultivation, overcoming seasonal limitations and ensuring a consistent supply of produce regardless of external climatic conditions.

Moreover, aeroponics' scalability and versatility make it applicable across various settings, from small-scale urban farms to larger commercial operations. Its space-efficient design, coupled with the ability to stack or arrange plants vertically, allows for higher crop density within limited areas, making it particularly suitable for urban agriculture. As technology and methodologies continue to advance, the potential for integrating aeroponics with other innovative practices, such as AI-driven monitoring systems or renewable energy sources, opens doors to further efficiency and sustainability in food production. This amalgamation of technology and agriculture heralds a future where aeroponics plays a pivotal role in meeting the growing global demand for fresh and nutritious food

4. KISAN SABHA

Another one of our propositions is the conduct of a Kisan Sabha, including farmers from Kanadia, Bijalpur, Jhalaria and Hingonia., held once every 3 months. This Kisan Sabha is to be made in collaboration with the Zilla Panchayat and the Zilla Panchayat CEO. It will aim at holding brief meeting with the small and marginal farmers to address the following:

- **The awareness about the technicalities of agro- economics or agricultural economics.**
- **The knowledge of new agricultural techniques for production.**
- **The introduction of new problems faced by the farmers to the authorities.**
- **The proposal of the formation of farmers' Self Help Groups, especially for mushroom farming and its propagation.**
- **Making the farmers aware about the government policies, schemes and subsidies to facilitate their production and its sale.**
- **Promoting the formation of consolidated funds of farmers living in the same locality for assistance in their production process. For example, funds for tractors, threshers, harvesters, tillers, trucks for transportation etc.**
- **Connecting the farmers to bankers to avail the facilities of loans, savings accounts, FDs, mutual funds, insurance etc.**

5. FORMATION OF SHGs

In a groundbreaking initiative poised to redefine the agricultural landscape, we are embarking on the establishment of Self-Help Groups (SHGs) for farmers. This pioneering endeavor aims to revolutionize the way farmers collaborate, learn, and thrive together. By creating a platform that empowers farmers through collective strength and shared knowledge, we envision a transformative impact on their socio-economic well-being. Through these SHGs, we aspire not only to enhance agricultural productivity but also to foster a sense of community, solidarity, and sustainable growth among farmers. This innovative approach heralds a new era, marking our commitment to shaping a brighter, more interconnected future for agriculture.

Creating self-help groups (SHGs) among farmers can be incredibly beneficial for their collective growth and support. Here's a step-by-step guide on how these groups will be formed:

Step 1: Identifying Potential Members

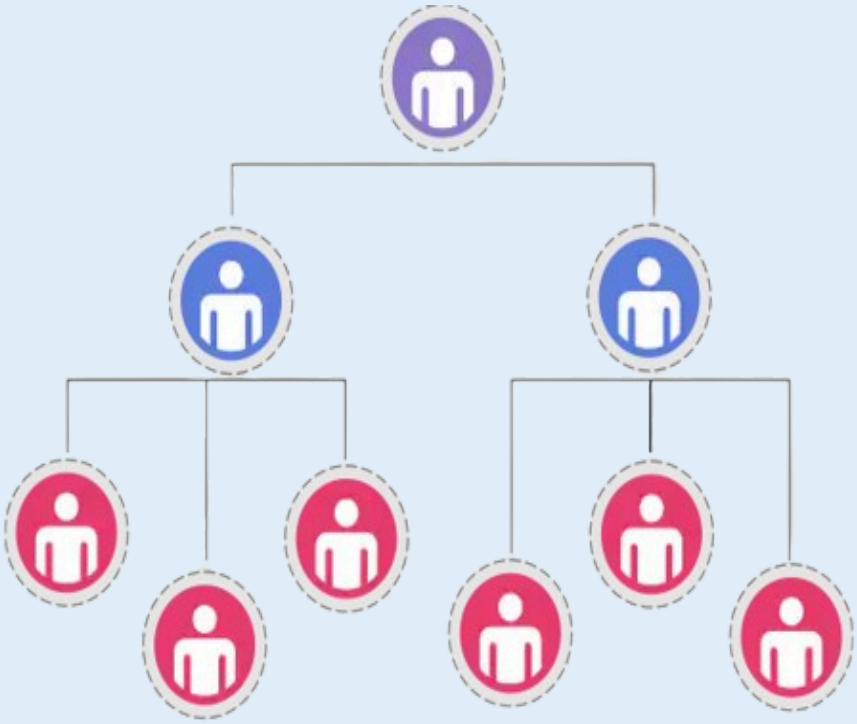
We will reach out to farmers in villages who are interested in joining these groups.
Aim for 10-15 members per SHG.

Step 2: Structuring the SHGs

We will organize meetings to explain the purpose and benefits of forming SHGs.
Each member contributes an amount between 500-1000 rupees as an initial deposit into a consolidated fund on a

Step 3: Establishing Oversight

A dedicated individual will be appointed to oversee each SHG within a village.
This person can be someone trusted within the community who has organizational skills or knowledge of financial management.



Step 4: Coordination among SHGs

The SHGs shall be grouped within a village together under a broader umbrella.
A Zonal head would be appointed at the district level to oversee and coordinate activities among these SHGs.

Step 5: Defining Objectives and Activities

Discussions would be encouraged within SHGs to determine their common objectives, whether it's collective purchasing, sharing resources, or knowledge exchange.
Regular meetings would be planned for sharing experiences, discussing challenges, and learning new techniques or information.

Step 6: Utilizing the Consolidated Fund

The consolidated fund can be used for various purposes like buying seeds in bulk, investing in equipment, organizing training sessions, or even providing financial assistance to members in need.

Step 7: Training and Capacity Building

Training sessions on modern farming techniques, financial literacy, and market trends to enhance the skills and knowledge of the SHG members would be arranged.

Step 8: Networking and Support

Encourage networking among SHGs, fostering a supportive environment where they can help each other in times of need or share successes and failures for mutual learning.

Step 9: Monitoring and Evaluation

The zonal head would regularly monitor the progress of SHGs and evaluate their activities to identify areas for improvement.

Adjust strategies and support based on the needs and challenges faced by the SHGs.

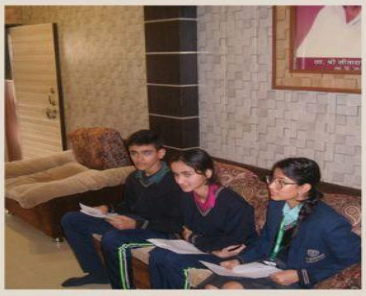
Step 10: Expansion and Sustainability

As SHGs grow and become more stable, their activities would be expanded and new SHGs would be formed in neighboring villages.

Sustainability would be encouraged by ensuring proper financial management and accountability within the groups.

Establishing self-help groups among farmers can significantly enhance their collective strength, knowledge, and financial stability. It's essential to foster an environment of trust, collaboration, and continuous learning within these groups to ensure their

SOCIAL INTERACTIONS



VISIT AT KANADIA

Aim:

The conversation aimed to uncover challenges faced by rural farmers: resource scarcity, lack of tech knowledge, unawareness of government aid, and uncertainties in market dynamics, seeking potential solutions.

The conversation shed light on the challenges faced by farmers in a rural setting. Despite their dedication to farming from a young age, insufficient access to resources and information remains a prevalent issue. The lack of high-end fertilizers affects crop yield, leading to disappointment in the outcome. New agricultural technologies like solar irrigation and automation systems are unfamiliar to them, as there's minimal exposure to or education about these advancements.

The disconnect between the Gram Panchayat's sporadic updates and the farmers' needs for comprehensive information about modern farming techniques exacerbates the problem. While government schemes like Kisan Credit Cards and hydroponics subsidies exist, farmers struggle to access or benefit from them due to the absence of guidance on application processes.

Their crop selection mainly revolves around wheat, soy, chickpeas, and corn, unaware of potentially profitable alternatives like mushrooms or lettuce. Despite a lack of knowledge, they express openness to new practices that promise increased profits, provided there's tangible proof of success.

Selling their produce directly at the Indore market minimizes intermediary involvement, but concerns about fair pricing persist. Overall, the farmers grapple with inadequate resources, limited exposure to technology, insufficient information on government schemes, and uncertainties about market dynamics, reflecting the complex challenges prevailing in their farming community.

VISIT AT BIJALPUR

Aim:

The aim was to explore and comprehend the multifaceted challenges faced by small farmers in rural Indore, highlighting issues of inadequate utilities, market access, technological awareness, and the absence of political support.

The conversation delved into the challenges faced by small farmers in rural Madhya Pradesh, featuring insights from Mr. Omprakash Rai and the farmers of Kanadia village. A major concern was the irregular provision of essential utilities like water and electricity by the government. Despite various schemes, small farmers felt excluded from benefits, which predominantly favored larger agricultural entities.

While the farmers showed interest in innovative techniques such as cultivating mushrooms, ayurvedic plants, and hydroponics, they encountered distinct obstacles. Access to markets through wholesalers for these crops posed a challenge, highlighting the necessity to eliminate middlemen to ensure fairer prices and stable income for farmers. Transportation hurdles further hindered their ability to reach markets effectively.

The absence of government officers' regular visits and limited awareness about modern farming technologies accentuated the farmers' difficulties. They expressed reluctance to adopt new methods due to fear and a lack of education. Additionally, the absence of regular farmer meetings for planning exacerbated their challenges.

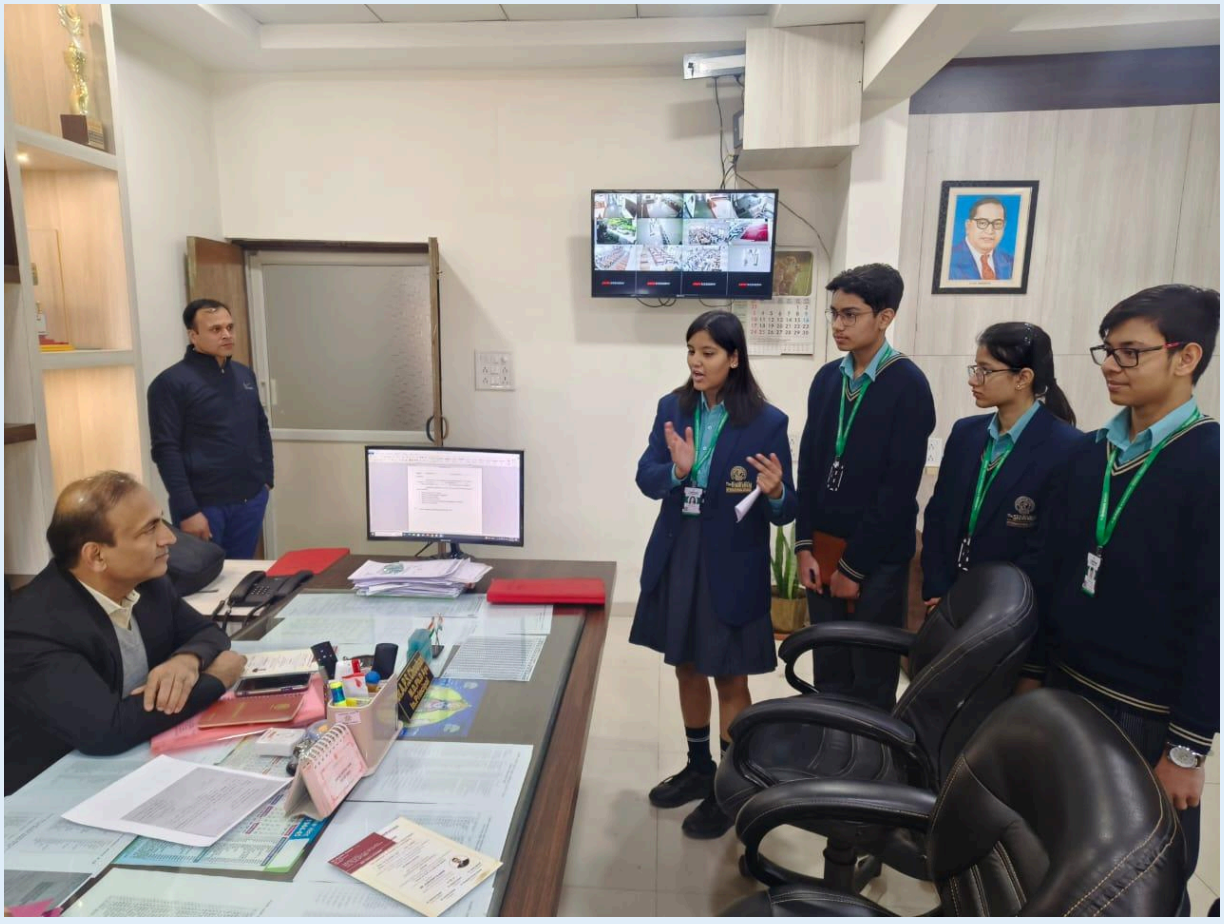
Political support was notably lacking, indicating a dearth of willingness to uplift small farmers. Key issues identified encompassed inconsistent utilities, transportation limitations, inadequate access to modern tools, middlemen exploitation, price instability, educational gaps, and insufficient political advocacy for small-scale farming.

VISIT AT THE ASHTANG AYURVED COLLEGE

AIM:

Our visit at the Government Autonomous Ashtang Ayurved College was to inquire about the growth, sale and usage of Ayurvedic and Herbal medicinal plants.

Through our visit at the college and our interaction with the administration of the college including the teachers, doctors and the administration of the college we were able to acquire important insights about Ayurvedic and Herbal medicinal plants. We came to know about the medicinal qualities of such plants along with their maintenance procedures and the practices that would be required to grow these plants.



INTERVIEW WITH MR. SHASHEESH SHRIVASTAV

AIM:

The interview aimed to explore the economic viability of oyster mushroom cultivation, highlighting its profitability, simplicity, government support, and potential integration into traditional farming for increased income.

In the interview focused on mushroom cultivation, the discussion revolved around the economic viability of oyster mushrooms, particularly highlighting their simplicity and affordability in cultivation. Emphasizing the lack of need for land, the conversation proposed a 2000 square foot mushroom house capable of yielding substantial harvests within a short span.

The profitability aspect was a key highlight, showcasing the potential for a 200% profit margin under optimal conditions. Government support in the form of schemes, subsidies, and training programs was also emphasized, indicating the accessible avenues for aspiring mushroom cultivators.
Personal experiences were shared, detailing a low-cost setup with significant yield, underscoring the accessibility of mushroom farming even with minimal investment. The interview explored the transformative impact of integrating mushroom cultivation into traditional farming practices, drawing inspiration from successful models in Himachal Pradesh and Delhi.

Furthermore, the conversation ventured into the prospect of utilizing vacant mines for mushroom cultivation, contingent on resource availability and specific conditions.

Overall, the interview painted a picture of mushroom cultivation as a feasible, profitable, and adaptable practice, shedding light on government aid and the potential for unconventional spaces to foster economic growth through agricultural innovation.

TECHNOLOGY EMPHASIZED

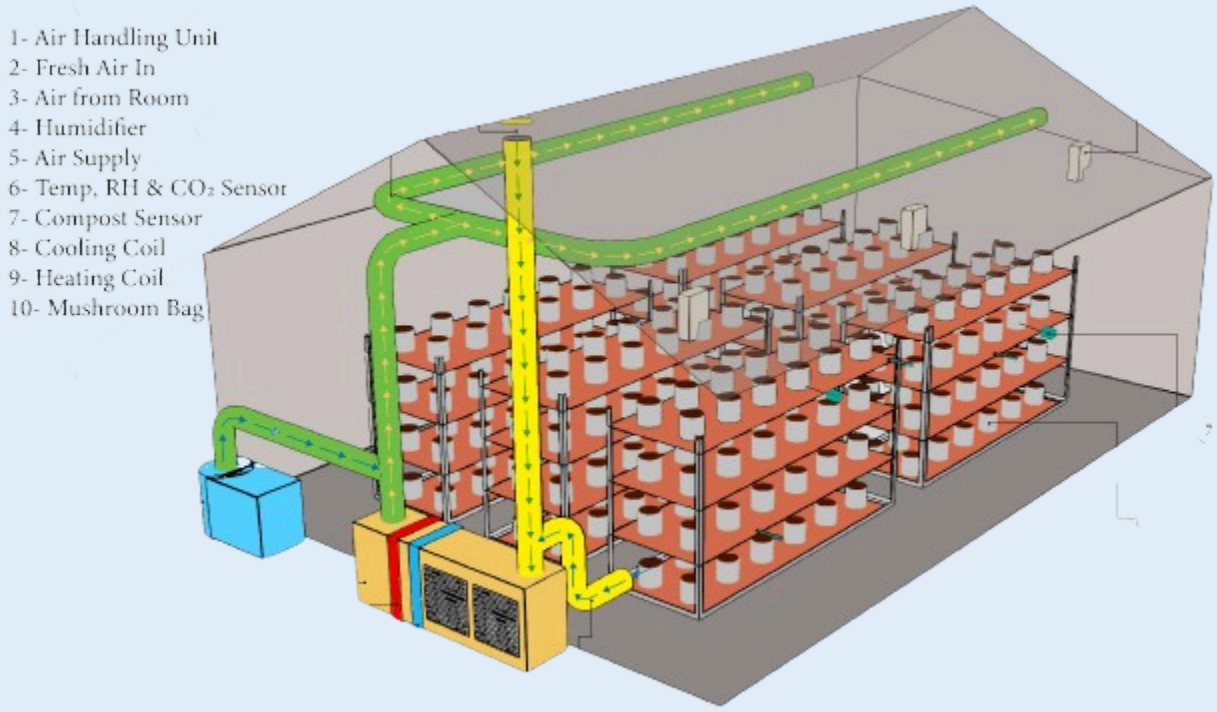
CULTIVATION OF MUSHROOMS IN MINES

- **Controlled Environment Agriculture (CEA): Mines can provide a controlled environment with stable temperatures and humidity levels. Controlled environment agriculture involves the use of technologies such as climate control systems, hydroponics, and artificial lighting to create optimal conditions for plant growth. This can be availed to by the farmers by linking up with government agencies, policies and private agents to an extent which are profitable for the farmers.Implementing automation technologies for tasks such as temperature and humidity control, irrigation, and monitoring can enhance efficiency and reduce the need for constant manual intervention.**
- **LED Lighting: In an underground environment, natural light may be limited. LED lighting systems can be used to provide the necessary spectrum and intensity of light required for mushroom growth. LED lights are energy-efficient and can be customized to match the light requirements of different stages of mushroom cultivation.**
- **Ventilation and Air Quality: Proper ventilation is crucial for maintaining air quality and preventing the buildup of carbon dioxide. It may be necessary to implement ventilation systems to ensure a healthy environment for mushroom growth. Properly designed and installed fans, inlets and controls minimize energy losses and ensure an environment that optimizes growth and comfort.**
- **Substrate and nutrient delivery system: In mushroom cultivation, substrate and nutrient delivery systems are crucial components for creating an environment conducive to the growth and development of mushrooms. Various organic materials can serve as substrates, including straw, wood chips, sawdust, agricultural by-products, and more. Substrates can be enriched with additional nutrients, such as bran or other supplements, to enhance mushroom growth.**
  Some cultivation methods involve liquid nutrient cultures. Mushroom mycelium is grown in a nutrient-rich liquid medium before being transferred to a solid substrate.

In advanced cultivation, hydroponic-like systems may be used, where nutrients are delivered in a liquid form to the mycelium. This is more common in laboratory or large-scale commercial setups.

For certain mushroom varieties, a casing layer is added on top of the substrate. It helps maintain moisture and provides a microenvironment suitable for fruiting. Casing layers can be made from peat moss, vermiculite, and other materials.

- **<u>Sustainable Practices:</u> Consideration should be given to sustainable practices, including waste management and recycling of materials used in the cultivation process. This is particularly important in an enclosed environment like a mine.**

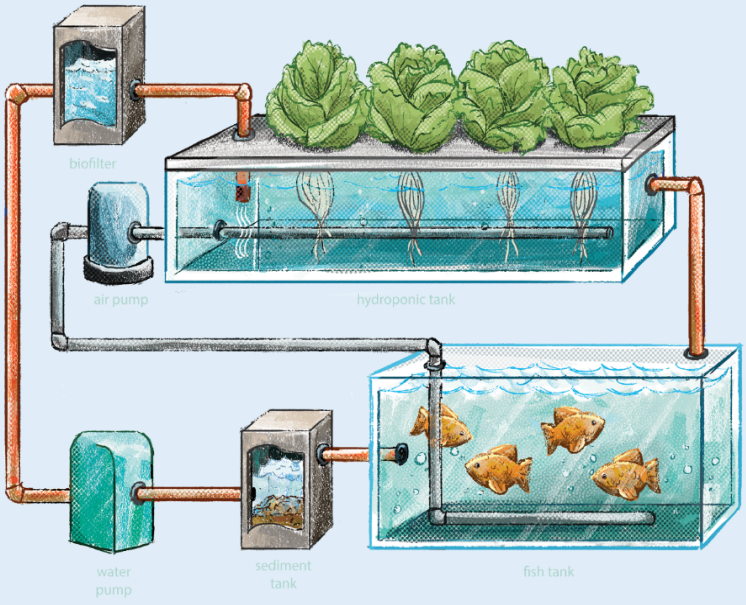


AYURVEDIC FARMING

- **Soil health: This involves the use of organic manures, compost, and natural fertilizers to enhance soil fertility. Crop rotation and intercropping are commonly practiced to prevent soil erosion and nutrient depletion.**
- **Natural pest control: Ayurvedic farming promotes the use of natural methods for pest and disease control. This includes the use of botanical extracts, neem oil, and other natural substances with pesticidal properties. Crop diversity is encouraged to create a natural balance that minimizes the risk of pest outbreaks.**
- **Companion planting: Companion planting is a common practice in Ayurvedic farming. Certain plants are grown together to provide mutual benefits, such as repelling pests or enhancing nutrient uptake.**
- **Biodynamic farming: Ayurvedic farming shares similarities with biodynamic farming, which involves a holistic approach to agriculture, considering the farm as a self-sustaining ecosystem. Biodynamic preparations, such as herbal teas and compost preparations, may be used to enhance soil vitality.**
- **Water Conservation: Efficient water management practices, such as rainwater harvesting and drip irrigation, are often integrated into Ayurvedic farming to conserve water resources.**

SOILLESS FARMING

- **Environmental Control: Automated systems can control and monitor environmental factors such as temperature, humidity, and light intensity. This ensures that plants receive optimal conditions for growth. Sensors placed within the hydroponic system collect real-time data, allowing for immediate adjustments to environmental parameters.**
- **Nutrient Delivery Systems: Automated nutrient delivery systems precisely control the composition and concentration of nutrient solutions. Pumps, timers, and sensors work together to deliver the right amount of nutrients directly to plant roots, promoting efficient nutrient absorption.**
- **pH and EC Monitoring: Automated systems can monitor and adjust the pH and electrical conductivity (EC) of the nutrient solution. Maintaining the correct pH level is crucial for nutrient absorption, and automation ensures continuous monitoring and adjustment as needed.**
- **Integrated Pest Management (IPM): Technology can be used to implement IPM strategies, such as the deployment of automated pest detection and control systems. Drones equipped with sensors can monitor crop health, and automated dispensers can release biological control agents or pesticides as needed.**



CONCLUSION

Our Organisation “KrishVeshan” focuses on farmers welfare using newly introduced innovative methods. Through our Research & Analysis and Social Interactions with the villagers we were able to identify the major problems faced by the agricultural community. After rigorous brainstorming sessions we were able to come up with a multitude of innovative solutions including development of proper communication channels, enhancing awareness and inculcating confidence with risk taking aptitude among farmers. We wanted to make the farmers aware about the schemes and practices related to their field as the Government appointed Gram Sevaks were not adequate and the Rural banks only worked to sell their loans. That is how we came up with the idea of establishing our own organization: “KrishVeshan” whose purpose is to organize farmers into SHGs and make them aware about schemes and practices beneficial to them through Kisan Sabhas and workshops by involving the dealers and NGOs. As we all know, agriculture is the backbone of the Indian economy with over 70% of the rural households relying on it for their livelihood. Without change being incorporated into this sphere, it is not far-fetched to say that the Indian economy might crumble under the demands of today and tomorrow. Thus, it is crucial that we stand for the farmers’ welfare, and move towards “Cultivating Prosperity Together”.

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